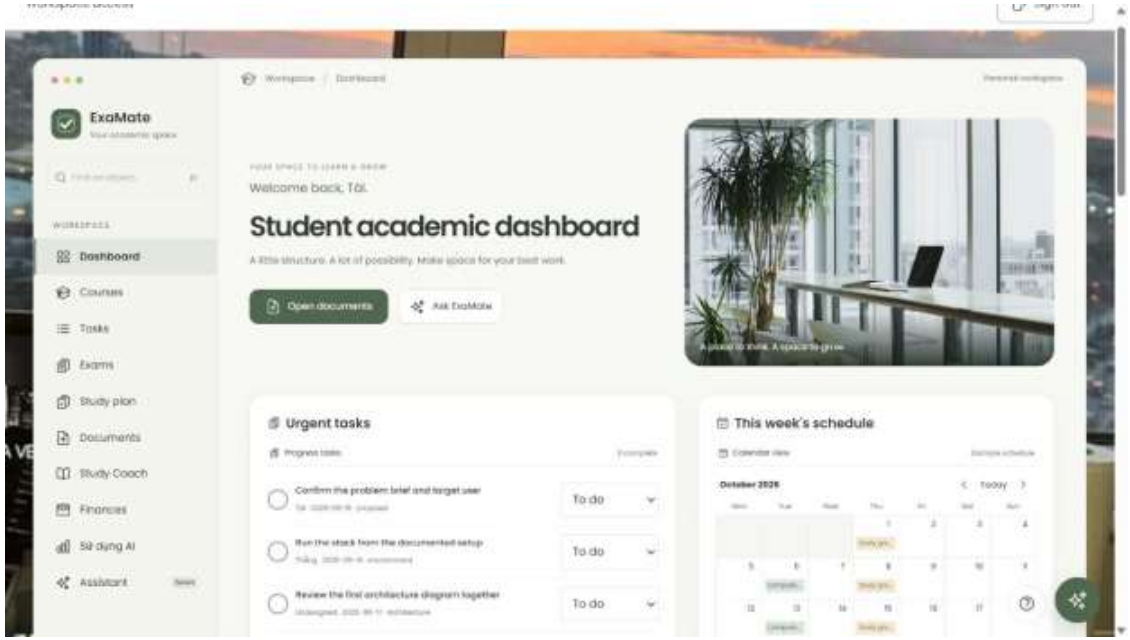
AI academic workspace

ExaMate

Study from your sources. Verify what AI says.

A · Grounded document Q&A

C · Structured study practice



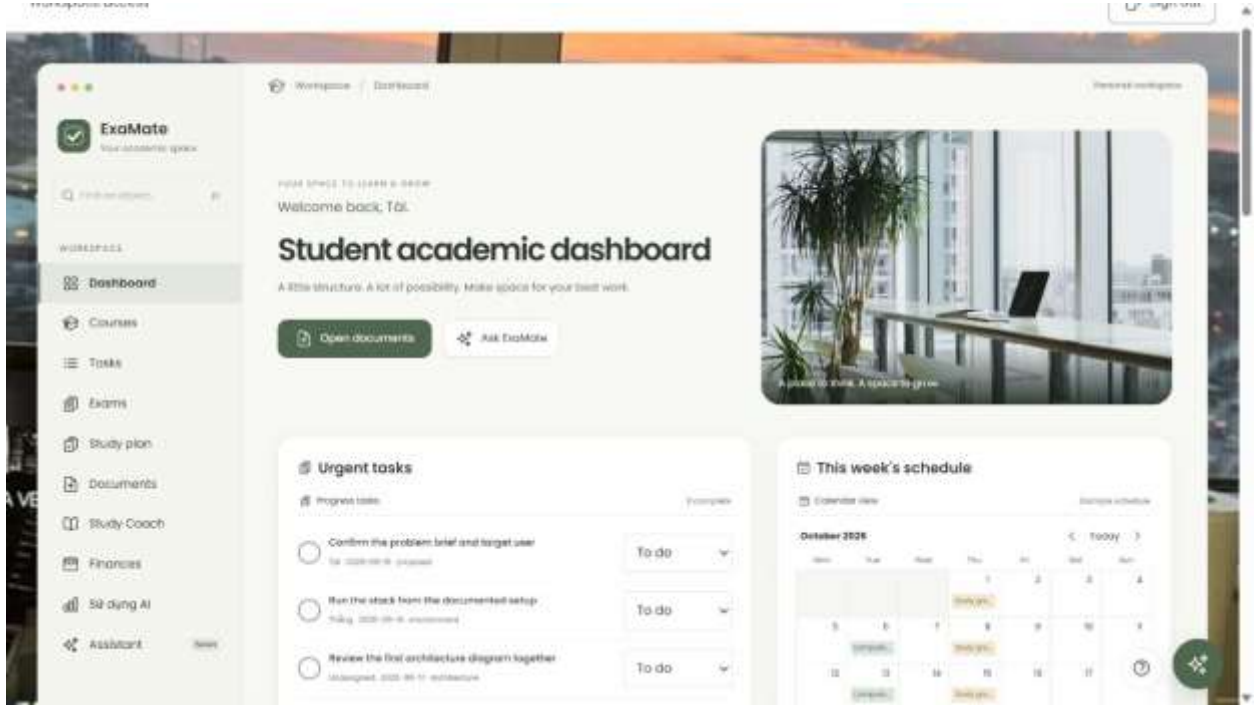
Tài + Thắng · New Technologies in Software Engineering · 2026–2027

From scattered files to a study loop

01 Organize
Tasks, courses, exams, plans, expenses and documents.

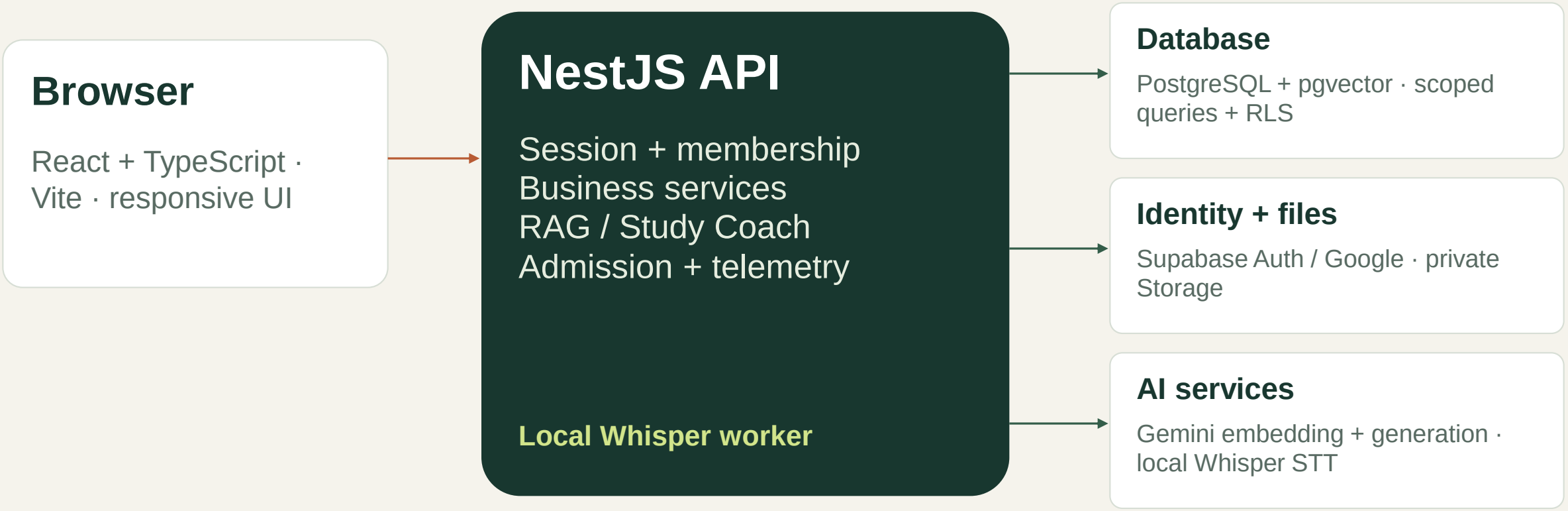
02 Verify
Ask within an authorized source; inspect the cited PDF page.

03 Practise
Preview, explicitly save, then review personal progress.



One workspace. A traceable learning loop. Real-service capture, retained.

Architecture with explicit trust boundaries



Business and Auth use separate non-owner runtime LOGINs.

Capability A: answer from authorized evidence

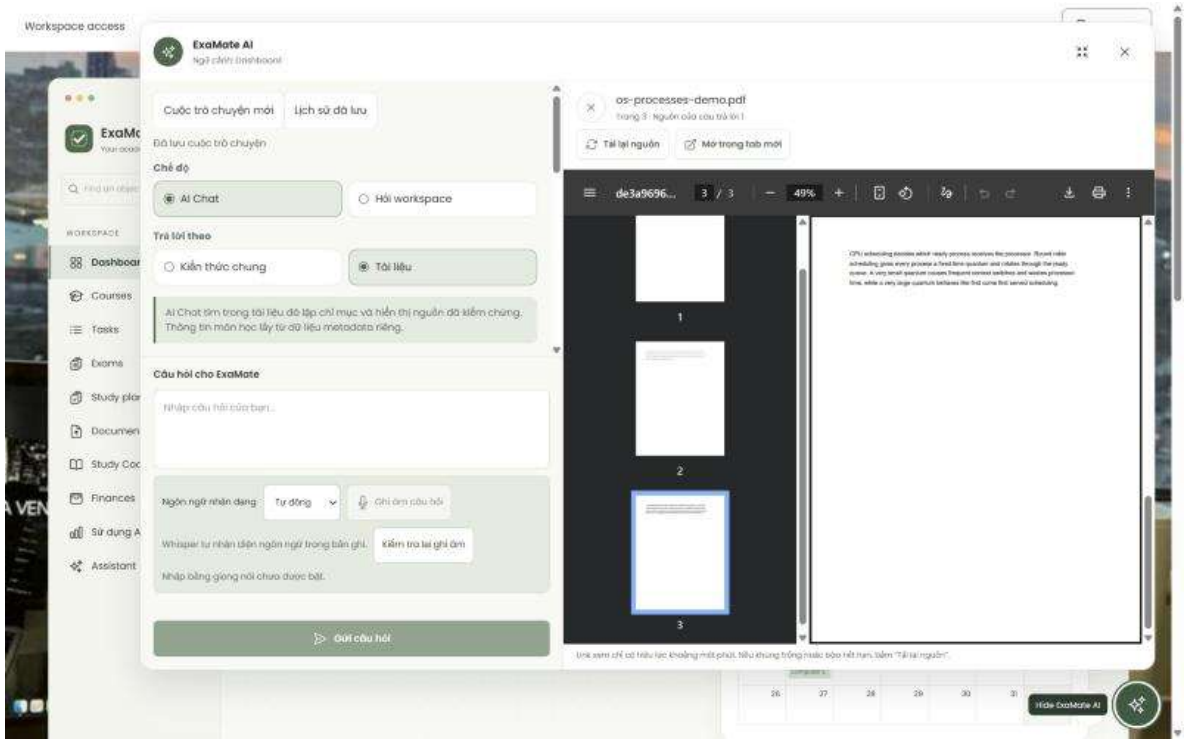
01 Ingest
Ready PDF → page-bounded chunks → 768-dimensional vectors.

02 Retrieve
Scope first; keep up to six qualifying evidence chunks.

03 Validate
Answer shape, source IDs and inline markers must agree.

04 Inspect
Open the cited PDF page with fresh scoped access.

No evidence → no answer generation.
Query embedding may already have run.



Historical browser fixture: real Nest/PG17; synthetic Auth, Storage and provider. PDF page 3 shown.

Live demo: verify one answer

01 Persist
Update a task, reload and confirm the stored change.

02 Ask
Select the ready operating-systems PDF and ask the question below.

03 Verify
Open page 3; then ask an out-of-document question.

04 Practise
Quiz preview → confirm → submit → progress; flip a saved card.

“What happens when the Round Robin time quantum is very small?”

http://127.0.0.1:8080 ↗ OPEN THE LOCAL DEMO

Capability C: preview becomes usable practice

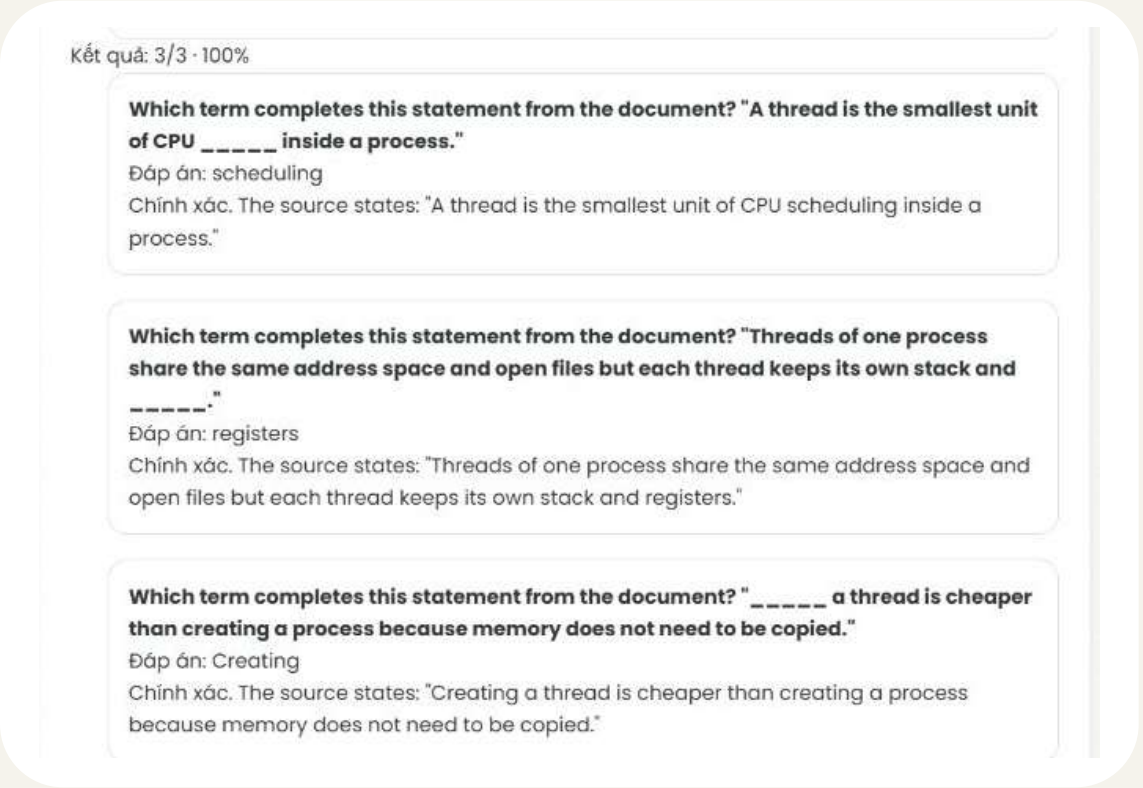
01 Preview
Ready document · count/language · evidence-bound JSON.

02 Confirm
Explicit save of a server-issued draft; private immutable snapshot.

03 Quiz
Server grades stored answers; attempts and progress persist.

04 Flashcards
Flip → self-rate → explicitly save a completed review.

Practice and progress do not call AI.



Historical fixture: real server grading, synthetic provider. Self-rating is not an AI grade.

Streaming and local voice, under user control

SSE / LIVE RESPONSE

General chat streams text; Stop aborts and ignores late output.

Stream status; publish the answer after source checks.

STT / LOCAL WHISPER

Record → transcribe
Edit → insert → send

60 s audio / 120 s inference
150 s request / 180 s outer wait

Physical microphone and Vietnamese accuracy still need acceptance. Speech input does not make the whole application offline.

Guardrails protect access and uncertain outcomes

SCOPE

Server principal + current membership + RLS; never trust a client role.

EVIDENCE

Treat document text as untrusted; refuse unsupported answers.

VALIDATION

Reject malformed structure or invented sources; no silent repair loop.

RECOVERY

Keep submission identity after uncertain writes; reconcile safely.

AI failure is visible; independent non-AI screens remain usable.
Injection resistance is inconclusive without the exact model context.

Verification: identify what each layer proves

2,324

APPLICATION TESTS

API Vitest 460 + Node 1,317
Web 547

521/521

CANONICAL PG17

Managed-profile fixture 1/1
reported separately

45

REAL HTTP CHECKS

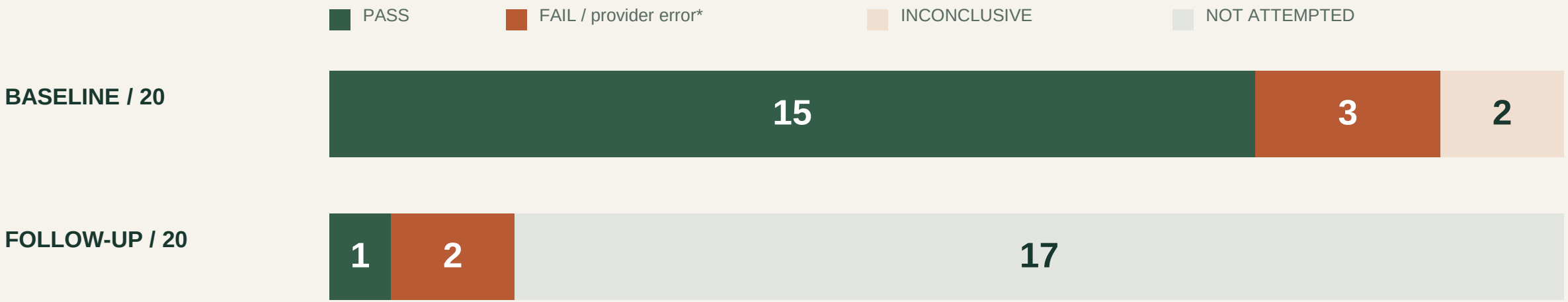
Docker + managed026
Google login observed

Browser integration: real Nest/PG17, synthetic upstreams for selected workflows. Real-service evidence is identified separately.

These are 23 October records, not fresh execution for this export.

Live evaluation: baseline and incomplete follow-up

Same model, corpus and prompt. Retrieval threshold: 0.55 → 0.50.



*Baseline: 3 contract failures. Follow-up: 2 provider HTTP503 errors.

A complete before/after quality claim is not yet supported.

AI-assisted review. Injection: INCONCLUSIVE. Semantic-content 18/20 is a separate measure.

Cost and latency: known values, stated limits

4.58 s

MEAN OPERATION

20 baseline operations

5.06 s

SAMPLE p95

Same 20; not a production SLA

$0.0128272

KNOWN GENERATION ESTIMATE

20 generations with usage

20 of 40 baseline calls had missing usage. Unknown is not zero.
Campaign upper used/reserved: USD 0.618895 across 51 calls, including a separate probe.

Telemetry is best-effort. Estimates are not invoices or a provider hard spending cap.

Two contributors, one integrated product

Tài

Final integration, frontend/business screens, Auth/RLS, RAG, Study Coach, AI controls and evaluation.

Thắng

Early backend/Supabase, initial Gemini integration, local STT and SSE.

Open: complete follow-up evaluation; broader semantic/device checks; personal confirmations.

Thank you. Let’s trace a request together.

Backup: frontend request lifecycle

UI

Components render views; hooks own state/effects; api.ts centralizes requests.

Identity

Principal/context query keys prevent old private data flashing back.

Request

Debounce → bounded GET → server page/count and scoped lookup.

Race guard

Abort + revision/key checks reject late responses.

Shared .d.ts catches compile-time drift, not runtime invalid data.

use-paged-search.ts · use-assistant.ts · api.ts · packages/contracts/index.d.ts

Backup: code and decisions to defend

RAG	document-chunker.ts → rag-retrieval.service.ts → assistant.service.ts
Study Coach	study-quiz-validation.ts · study-coach.service.ts · study-flashcards.service.ts
Voice + SSE	audio-inspector.ts · local-speech-runtime.ts · assistant-stream.ts
Auth + AI controls	auth.guard.ts · ai-admission.service.ts · ai-usage.service.ts

Show a function, explain one regression and name one remaining limit.